BT4222 Group 13: Dataset & Source Code Listing

Personalised Product Search with JDSearch (DHAR re-ranker)

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Item	Link
GitHub repository (branch main)	github.com/steffanylars/nus-web-mining-business-insights
Main README (structure, how to run, Colab links)	README.md
Notebooks (Steps 1 to 7)	Final Project/notebooks/
Source code (scripts)	Final Project/src/
Intermediary datasets	Final Project/data/intermediate/
Saved outputs and diagnostics	Final Project/artifacts/
Traceability of reported numbers	Final Project/TRACEABILITY.md
Raw dataset (official source)	github.com/rucliujn/JDsearch

1. Datasets

Link: https://github.com/steffanylars/nus-web-mining-business-insights/tree/main/Final%20Project/data/intermediate

Dataset 1: JDSearch raw files (user_behavior_data.txt, product_meta_data.txt)

- **Purpose:** Source data for the whole project.
- **Content:** Real JD search queries, candidate slates with click, cart and order labels, each user's preceding history with time gaps, and anonymised product metadata (about 5.8 GB).
- **Used in:** Steps 1 and 2. Not stored in the repository because of its size; download it from the official source linked above.

Dataset 2: Sampled modelling tables (candidates, history, sessions, meta_subset .parquet)

- **Purpose:** Cleaned and intermediary tables, so Steps 3 to 7 run without the raw files.
- **Content:** 20,000 users sampled with seed 42 and split by user 70/15/15 (14,000 / 3,000 / 3,000). meta_subset keeps only the product columns the models read.
- **Used in:** Steps 2 to 7.

Dataset 3: Feature table (features.parquet)

- **Purpose:** Engineered features for the rankers.
- **Content:** 26 numeric features per query and product pair: short-term intent, order history, query match and training-only popularity.
- **Used in:** Steps 3 to 7.

2. Source Code / Notebooks / Model Outputs

Link: https://github.com/steffanylars/nus-web-mining-business-insights/tree/main/Final%20Project/notebooks. Each notebook explains its step in markdown, re-checks the saved results and ends with a highlighted *Key results* cell. The code also runs in Google Colab (links in the README).

1. Step1_Temporal_Analysis.ipynb

- **Purpose:** Turns time gaps into event ages for all 26.7M history events and measures how long clicks, carts and orders are logged.
- **Output:** retention_cutoffs.txt, temporal tables and figures

2. Step2_Build_Dataset.ipynb

- **Purpose:** Samples 20,000 users, builds candidate and history tables and splits by user.
- **Output:** dataset_manifest.json (split fingerprint), sampled tables

3. Step3_Feature_Engineering.ipynb

- **Purpose:** Builds the 26 features.
- **Output:** features.parquet, feature_summary.json

4. Step4_Baselines_LambdaMART.ipynb

- **Purpose:** Random, popularity, content, SVD and LambdaMART rankers.
- **Output:** results_baselines_lgb.json, lgb_model.txt

5. Step5_Neural_Rankers_DHAR.ipynb

- **Purpose:** DHAR, matched neural controls and a three-seed check.
- **Output:** results_neural.json, multiseed_summary.csv

6. Step6_Diagnostics.ipynb

- **Purpose:** Ablation, paired bootstrap, subgroup analysis and DHAR attention analysis.
- **Output:** ablation.json, paired_bootstrap.json, dhar_attention_by_age_type.csv

7. Step7_ColdStart_Diversity.ipynb

- **Purpose:** History-free fallback routing and category diversification.
- **Output:** results_coldstart_diversity.json, diversity_tradeoff.csv

3. Other folders and files

Folder: src/

- **Purpose:** The same code as plain Python scripts (the notebook cells are cut from these files), plus run_pipeline.py to run all steps in order.

Folder: artifacts/corrected/

- **Purpose:** Metrics, predictions and figures of the final run. Every number in the report comes from this folder.

Folder: artifacts/step7_corrected/ and artifacts/diagnostics/

- **Purpose:** Step 7 results on the corrected features, the three-seed summary and the attention table and figure.

Folder: artifacts/archive/

- **Purpose:** Superseded working files, kept only for history.

Folder: figures/

- **Purpose:** End-to-end pipeline diagram (fig_pipeline.png).

Folder: tests/

- **Purpose:** Regression tests and a check that notebooks and scripts stay in sync.

File: TRACEABILITY.md

- **Purpose:** Each reported number, the file it comes from and how to recompute it.

File: requirements.txt

- **Purpose:** Python package versions.